

Shivam & Shantanu 5-Day Sprint — Step-by-Step (Shivam = infra/token/DEX lead, Shantanu = bridge/wallet/exchange integration lead)

Before Day 1 — preflight (do this immediately)

1. Sync codebase
 - git checkout main && git pull origin main
 - Create sprint branch: git checkout -b sprint/mainnet-ready
2. Backup current state and DB snapshots.
3. Ensure access to cloud/VMs, Docker registry, monitoring credentials, and a secure place for keys (.env not committed).
4. Create a shared sprint folder (e.g., sprint-mainnet/) for artifacts.

MVPD (preflight): push sprint/mainnet-ready branch and create a Sprint README listing hosts and contact persons. Submit short reflection.

Day 1 — DEX Hardening + Multisig OTC

Goal: enforce slippage protection, run edge-case swap tests, implement 3/5 multi-sig OTC override.

Shivam (primary)

1. Add slippage protection to pool contract/back-end swap logic:
 - Enforce minAmountOut pattern: caller must supply minAmountOut, require amountOut >= minAmountOut. (If on-chain: require(amountOut >= minAmountOut, "slippage");).
 - Add maxDelta check: require (newPrice - oldPrice)/oldPrice <= maxDelta or reject.
2. Add a server-side emergency circuit breaker: if price impact > threshold, reject or route to multisig approval queue.

Shantanu (support)

1. Prepare test harness script scripts/dex_slippage_test.sh:
 - Create pool with tiny reserves (e.g., 100 tokenA / 1 tokenB),
 - Attempt a very largeswap,
 - Validate that tx reverts or minAmountOut protection triggers.

Multisig OTC (both)

1. Deploy or reuse a multisig wallet (3-of-5):
 - If Solidity/EVM: use a standard Gnosis Safe or a simple multisig wallet contract. If non-EVM, implement server-side multisig signing (store signers, require 3/5 signatures).
2. Implement OTC flow:
 - requestOTC(tradePayload) → stores pending trade with txHash and requiredSignatures = 3.
 - approveOTC(tradeId, signerSig) → collect signatures.
 - When signatures ≥ 3 , execute or release escrow.
3. Test override:
 - Simulate OTC fail, require admin override via 3 signers to release funds.

Acceptance criteria (Day1):

- Unit tests and script show slippage protection triggers for edge-case swaps.
- OTC request + 3-of-5 override flow tested successfully.
- Artifacts: scripts/dex_slippage_test.sh output, multisig contract address (or server impl), logs.

Submit at EOD:

- test logs/screenshots, PR to sprint branch, and reflection (3–5 sentences).

Day 2 — Bridge + Wallet Round-trip + Full Logs

Goal: complete a full wallet→DEX→bridge→other-chain roundtrip with full event trail.

Shantanu (primary)

1. Finalize bridge retry/dedup + logging endpoints:
 - POST /relay/eth and POST /relay/sol (accept signed bridge messages).
 - Ensure dedupe: reject duplicate eventHash.
 - Write relay proof: every relayed event writes relayLog{eventHash, srcChain, dstChain, txHash}.
2. Provide CLI or script scripts/roundtrip_test.sh:
 - Step 1: wallet signs & submits swap to DEX,
 - Step 2: DEX emits event → bridge picks and relays to TargetChain,
 - Step 3: TargetChain receives mint/unlock confirmation.

Shivam (support)

1. Ensure token approvals/log emissions are in expected format:
 - event fields: {sourceChain, token, amount, sender, receiver, txHash, timestamp}.
2. Confirm wallet RPC endpoints are configured and accessible from the relay node.

Acceptance criteria (Day2):

- scripts/roundtrip_test.sh completes and writes a roundtrip-proof.json containing all event hashes and tx hashes across chains.
- /log/events shows the full trace for the test.
- Artifacts: roundtrip-proof.json, relay logs, console output.

Submit at EOD: roundtrip logs/screenshots, PR, reflection.

Day 3 — Staged Mainnet Deploy + Explorer + Faucet

Goal: bring up mainnet nodes (staged), deploy DEX with safety logic, get explorer showing txs; set up faucet for liquidity seeding.

Shivam (primary)

1. Genesis + chain-id:
 - Finalize genesis.json (supply cap, allocations).

- Start at least 3 validator nodes (on separate VMs or containers).
 - `docker-compose -f deploy/mainnet/docker-compose.yml up -d`
2. Deploy DEX contracts (with slippage + OTC multisig).
 3. Deploy token contract (or set native token parameters).
 4. Setup faucet service for test liquidity (rate limited, authenticated).

Shantanu (primary for explorer)

1. Deploy a minimal explorer or expose block/tx API:
 - Option A: lightweight custom explorer (REST endpoints: `/block/:height`, `/tx/:hash`).
 - Option B: configure an existing explorer if chain is EVM compatible (Blockscout). If not EVM, a simple indexer script that writes blocks/txs into a DB and a small UI is OK.
2. Verify explorer shows the DEX transaction.

Acceptance criteria (Day3):

- Mainnet nodes are producing blocks.
- Explorer returns block and tx info for the DEX tx from Day2.
- Faucet can fund a wallet and transaction succeeds with visible block confirmation.

Submit at EOD: node logs, explorer screenshots, faucet usage log, reflection.

Day 4 — Exchange Readiness (Docs + Compliance + Listing bundle)

Goal: produce the package exchanges require + seed DEX liquidity and test listing flows.

Shivam & Shantanu (joint)

1. Prepare Listing Bundle (ZIP) containing:
 - Token spec: name, ticker, decimals, totalSupply, circulatingSupply, contractAddress, deployTxHash.
 - Tokenomics doc: cap, emission schedule, vesting (if any).
 - Security: audit summary (list any audits, or a minimal internal audit checklist showing no hidden mint functions).
 - Explorer links, RPC endpoints, block confirmation times.

- KYC/AML statement (if required).
- 2. Seed DEX liquidity:
 - Create pair BHC/BHT or BHC/Stable depending on DEX.
 - Seed initial liquidity from the faucet/admin wallet.
- 3. Submit listing to one DEX (AMM) and prepare DEX listing PR or contact information for CEX (if pursuing CEX, follow their submission portal steps).
- 4. QA: run swap with seeded liquidity and verify slippage protection works.

Acceptance criteria (Day4):

- Listing bundle ZIP is complete and committed to sprint folder.
- Liquidity pool seeded and shows in explorer & DEX UI.
- Documentation includes step-by-step listing instructions.

Submit at EOD: listing bundle ZIP, liquidity tx hashes, DEX pool screenshot, reflection.

Day 5 — Monitoring, Stress Test, Demo & Handoff

Goal: full stress test, monitoring dashboards, final demo video and clear handoff docs.

Shivam (primary)

1. Monitoring:
 - Prometheus metrics collector (expose endpoints for block_height, tx_latency, peer_count, pool_health).
 - Grafana dashboard with charts for TPS, p50/p95/p99 latency, pool health, bridge retry rate.
 - Alerts: Slack/Email rule for block_time > 3x expected and DLQ > threshold.
2. Stress test:
 - Run scripts/stress_test.sh (simulate 1k+ txs across wallet/DEX/bridge).
 - Measure errors, memory, CPU, and record graphs.

Shantanu (primary)

1. Demo & handoff:

- Record 5–8 minute demo: wallet → seed liquidity → swap → bridge → explorer → Grafana.
- Write mainnet-deploy.md with steps to reproduce (one-command deploy if possible).
- Tag release: `git tag v1.0-mainnet && git push origin --tags`

2. Acceptance criteria (Day5):

- Stress test completed with acceptable failure rate (documented).
- Grafana dashboard live with historical stress test graphs.
- Demo video + mainnet-deploy.md submitted.

Submit at EOD: stress test logs, Grafana screenshots, demo video link, mainnet-deploy.md, final reflection.

Deliverables checklist (every day)

- PR(s) to sprint/mainnet-ready with code + tests.
- Logs/screenshots proving MVPD criteria.
- docs/reflections/{shivam,shantanu}/dayN.md — 3–5 sentence reflection covering Humility, Gratitude, Honesty/Integrity.
- One consolidated ZIP at sprint end: sprint-mainnet-artifacts.zip (genesis, contracts, explorer DB dump, monitoring screenshots, demo video link, listing bundle).

Quick commands & snippets (copy/paste)

- Build & run node locally in Docker:

```
git checkout sprint/mainnet-ready
docker build -t blackhole/node:dev .
docker run --env-file .env -v ./data:/data -p 26656:26656
blackhole/node:dev
```

- Run dex slippage test:

```
bash scripts/dex_slippage_test.sh
# check outputs: logs/slippage_test.out
```

- Run roundtrip bridge test:

```
bash scripts/roundtrip_test.sh
# outputs: roundtrip-proof.json
```

- Start Prometheus + Grafana:

```
docker-compose -f deploy/monitoring/docker-compose.yml up -d
# visit Grafana at http://<host>:3000
```

Safety & rollback

- Keep an automated snapshot step before each major deploy:

```
# quick DB snapshot example
tar czf data-snapshot-$(date +%s).tgz ./data
```

- If a mainnet change breaks consensus, bring validators to maintenance mode and rollback to last working genesis/config snapshot.

Scoring & acceptance (final)

- Each day scored by: Technical (10), Values (15), Discipline/timely submission (5) → /30 per day.
- Final sprint score = average of 5 day scores. Submit final self-eval + reviewer comments.